A Bayesian WeightWatcher: Calibrated Heavy-Tailed Spectral Diagnostics and What They Overturn

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Paper under double-blind review

Abstract

Heavy-Tailed Self-Regularization (HTSR) reads the quality of a trained neural network off the eigenvalue spectrum of its weight matrices: the power-law tail exponent α is a label-free generalization proxy, and Martin's renormalization-group theory sharpens this into the prediction that $\alpha = 2$ is a robustness-optimal fixed point. The reference tool, WeightWatcher, reports α as a single point estimate from one Kolmogorov–Smirnov scaling window, and never tests whether a layer is a power law at all. We show this matters. We introduce wwj/wwjd, a JAX/Equinox reimplementation that adds a calibrated Bayesian layer: a closed-form Gamma–Pareto posterior over α, Bayesian model averaging over the scaling window, Bayesian model comparison against exponential, lognormal, truncated power-law, and generalized-Pareto tails, posterior-predictive checks, a hierarchical cross-layer posterior, and an errors-in-variables regression that turns α into a calibrated accuracy predictor. Reanalyzing WeightWatcher's own canonical checkpoints, we find: the famous "GPT2 is better trained than GPT" gap largely dissolves once the scaling window is marginalized (mean α 4.12 $\rightarrow$ 2.90 for GPT, the GPT–GPT2 gap shrinking threefold), with two independent goodness-of-fit tests contradicting the "not a power law" mechanism and showing the inflation is truncated-power-law misspecification; the flagship VGG accuracy trend is confirmed and sharpened under the posterior ($r{=}-0.77 \rightarrow -0.98$); and batch normalization breaks the power law in 8–17% of layers, invisible to the point estimate. Training GPT-2 ourselves, we further isolate that α tracks training maturity, not data domain: a one-epoch domain fine-tune leaves the spectrum untouched while a one-epoch from-scratch model sits far from $\alpha = 2$ (heavy tail not yet formed). We then apply the same diagnostics across modalities, where the $\alpha = 2$ robustness prediction holds for physics-realistic neuroimaging perturbations but fails to transfer to a foundation-model-scale pathology intervention. We release wwj as an open tool for uncertainty-aware spectral diagnostics.

1 Introduction

The Heavy-Tailed Self-Regularization (HTSR) framework (Martin & Mahoney, 2021a; 2019) observes that the empirical spectral density (ESD) of weight matrices in well-trained deep networks develops a heavy, approximately power-law tail $p(\lambda) \propto \lambda^{-\alpha}$, and that the exponent α predicts generalization across hundreds of pretrained models with no access to data or labels (Martin et al., 2021). Martin's renormalization-group (RG) theory of learning (Martin, 2026) and its semi-empirical successor (Martin & Mahoney, 2025) sharpen this into a quantitative claim: $\alpha = 2$ is a critical fixed point, and proximity to it should confer robustness.

These are strong, falsifiable claims, and they rest entirely on how α is estimated. The reference tool, WeightWatcher, fits α by maximum likelihood on the tail above a single scaling-window lower bound $x_{\min}$ chosen to minimize a Kolmogorov–Smirnov (KS) distance, and reports a per-layer point value. This pipeline has two well-known soft spots that the authors themselves flag: the MLE "works very well for $\alpha \in (2, 4)$...adequate, although imprecise, for smaller and especially larger α" (Martin et al., 2021), and the KS objective sometimes has no well-defined minimum in $x_{\min}$ (Martin & Mahoney, 2021b); the tool ships a truncated-power-law option precisely because plain power-law fits over-estimate α. Yet downstream conclusions—"this model is poorly trained because it has unusually large α," "this layer is ideal because $\alpha \approx 2$"—are read off the point estimate as if it carried no uncertainty, and the power-law assumption underlying α is never tested.

We argue that a Bayesian treatment is not a cosmetic addition but changes published conclusions. We introduce wwj, a JAX/Equinox (Bradbury et al., 2018; Kidger & Garcia, 2021) reimplementation of WeightWatcher, and wwjd, a Bayesian layer on top of it (Section 3): a closed-form conjugate posterior over α, Bayesian model averaging (BMA) over $x_{\min}$, Bayesian model comparison that adds the truncated power law and the extreme-value generalized Pareto to the usual exponential/lognormal alternatives, posterior-predictive goodness-of-fit, a hierarchical cross-layer population posterior, and an errors-in-variables (EIV) regression that propagates α uncertainty into a calibrated accuracy predictor, together with region-of-practical-equivalence and prior-sensitivity diagnostics.

Our contributions are: (1) the wwj/wwjd tool, including exact CSN $x_{\min}$ selection (Clauset et al., 2009), a differentiable Hill estimator enabling direct spectral regularization, and the Bayesian model-criticism suite; (2) a reanalysis of WeightWatcher's own canonical checkpoints (Section 4) showing that the GPT/GPT2 "training quality" gap is largely a scaling-window artifact, that the flagship VGG accuracy trend strengthens under the posterior, and that batch normalization silently breaks the power law; and (3) a cross-modality application (Section 5) where the $\alpha = 2$ robustness prediction holds for realistic neuroimaging perturbations but does not transfer to a pathology foundation-model intervention—an honest map of where the theory does and does not pay off.

2 Related Work

HTSR and spectral quality metrics. Martin & Mahoney (2021a; 2019) established heavy-tailed weight spectra and the "5+1 phases" of training; Martin et al. (2021) showed α predicts test-accuracy trends without data. The capacity metric is the weighted alpha $\hat{\alpha} = \mathrm{mean}_\ell \alpha_\ell \log_{10} \lambda_{\max,\ell}$. The RG grounding is Martin (2026); Martin & Mahoney (2025); the contest post-mortem (Martin & Mahoney, 2021b) documents a Simpson's paradox between shape (α) and scale metrics. All of these read α as a point estimate; we put a posterior on it.

Power-law fitting and its critique. The CSN MLE-plus-KS procedure (Clauset et al., 2009) and the powerlaw package (Alstott et al., 2014) underlie HTSR fitting. The statistical risks—$x_{\min}$ instability, alternative heavy-tailed distributions, finite-size truncation—are well documented, and motivate Bayesian model averaging over $x_{\min}$, model comparison, and the truncated power law and generalized Pareto from extreme-value theory (Coles, 2001) that we add.

Bayesian workflow and model criticism. Power-scaling prior-sensitivity (Kallioinen et al., 2024) and predictive model comparison (Vehtari et al., 2017) are standard tools for trustworthy Bayesian conclusions; we adapt them to spectral diagnostics.

Spectral shaping and robustness. The Muon optimizer (Jordan et al., 2024) is conjectured to shape spectra toward the RG optimum; our differentiable alpha_loss is a direct, optimizer-agnostic alternative. For the cross-modality application we use TorchIO (Pérez-García et al., 2021) perturbations, the MONAI (Cardoso et al., 2022) ViT, and the BrainIAC (Pati et al., 2024) and MindEye (Scotti et al., 2024) checkpoints.

3 Method: wwj and wwjd

Frequentist core (wwj). For each 2D weight matrix W with min(shape) ≥ 50 we compute the eigenvalues of $W^\top W/N$ and select $x_{\min}$ by exhaustive KS minimization over every eigenvalue as a candidate (matching CSN 2009, rather than the reference's log-spaced grid).

A differentiable Hill estimator $\hat{\alpha} = 1+n(\sum_i \log \lambda_i/x_{\min})^{-1}$ backpropagates through eigvalsh, enabling the regularizer $\mathcal{L}_\alpha = \lambda_{\text{reg}} \cdot \text{mean}_\ell(\hat{\alpha}_\ell - 2)^2$. A Vuong likelihood-ratio test compares power-law against exponential and lognormal tails.

Bayesian layer (wwjd). The tail above $x_{\min}$ is Pareto, so $t = \log(\lambda/x_{\min})$ is exponential with rate $\beta = \alpha - 1$, and a conjugate Gamma(a_0, b_0) prior on β yields the closed-form posterior $\beta \mid \text{data} \sim \text{Gamma}(a_0 + n, b_0 + \sum t)$. Hence the posterior mean, credible interval, and $P(\alpha < 2)$ are exact, and the flat-prior limit recovers the CSN MLE. On top of this single-window posterior we build: (i) BMA over $x_{\min}$, a Gamma mixture weighting each candidate window by a comparable marginal likelihood, with a law-of-total-variance decomposition that separates within-window from window-choice uncertainty; (ii) Bayesian model comparison with analytic evidences for power-law/exponential/lognormal and Laplace evidences for the truncated power law and generalized Pareto; (iii) a posterior-predictive p-value from Lomax replication; (iv) a hierarchical NumPyro population posterior over per-layer α with shrinkage; (v) an errors-in-variables regression of a downstream metric on $\hat{\alpha}$ that propagates the $\hat{\alpha}$ posterior SD as measurement error; and (vi) ROPE ($P(\alpha \in [2-\delta, 2+\delta])$) and power-scaling prior-sensitivity diagnostics. The conjugate per-layer surfaces are pure JAX; only the hierarchical model samples.

4 Validation on WeightWatcher's Own Canonical Cases

Before applying the Bayesian diagnostics to new modalities, we test them where the answer is already published: on the exact pretrained checkpoints Martin & Mahoney used to establish heavy-tailed self-regularization. We load each model from its public source (torchvision ImageNet weights (Simonyan & Zisserman, 2015); HuggingFace GPT/GPT2 (Radford et al., 2019)), extract every 2D weight matrix with min(shape) $\geq$ 50, and compute both the frequentist WeightWatcher point estimate and the wwjd posterior (Table 1). Three findings result.

The "GPT2 is better trained than GPT" gap largely dissolves under $x_{\min}$ marginalization. The canonical NLP example holds that GPT is poorly trained because it has "numerous unusually large α," while GPT2-small has all $\alpha \leq 6$. Under the frequentist single-$x_{\min}$ estimate we reproduce this: mean $\alpha = 4.12$ for GPT versus 3.62 for GPT2 and 3.70 for GPT2-medium. Under the wwjd posterior that marginalizes over the scaling window, all three collapse into one healthy band: 2.90, 2.73, and 2.75 respectively (Figure 1). The frequentist gap (GPT − GPT2 = 0.50) shrinks 3.1× to 0.16. Per layer, GPT's 9 768 × 768 attention-projection matrices fall from frequentist $\alpha \in [5.2, 6.9]$ to posterior $\alpha \in [2.8, 3.5]$. The shift is not an estimator artifact: 18–35% of the per-layer α variance on these layers is attributable to the $x_{\min}$-window choice alone (the law-of-total-variance decomposition of the posterior), exactly the degree of freedom the single KS-optimal window collapses.

Two independent goodness-of-fit tests then contradict the "not well-described by a power law" mechanism on these same layers. The Bayesian model comparison prefers a power law over both exponential (log Bayes factor +120 to +163) and lognormal tails, and the posterior-predictive p-values are all above 0.05 (0.26–0.68): the power-law fit is adequate in absolute terms. When we extend the comparison set to the truncated power law—the model WeightWatcher itself ships because plain power-law fits over-estimate α—and the generalized Pareto (the extreme-value peaks-over-threshold tail), the truncated power law beats the plain one on all 50/50 GPT layers. The "large α" is therefore a power-law-with-truncation mis-specification, made explicit: when the finite-size cutoff is modeled, the inflation disappears (Figure 3). A region-of-practical-equivalence test keeps this honest—only 1/50 GPT layers place majority posterior mass in $\alpha \in [1.75, 2.25]$, so the Bayesian $\alpha \approx 2.90$ is far below the frequentist 4.12 but still above the RG-ideal 2.0. A power-scaling prior-sensitivity diagnostic confirms the conclusions are not prior-driven (mean sensitivity 0.001 posterior-SD units).

The VGG weighted-α trend is confirmed—and sharper under the posterior. WeightWatcher's flagship claim is that the weighted alpha $\hat{\alpha} = \text{mean}_\ell\, \alpha_\ell \log_{10} \lambda_{\max,\ell}$ tracks test accuracy across the VGG11/13/16/19 (±BN) ImageNet series. We reproduce this with the

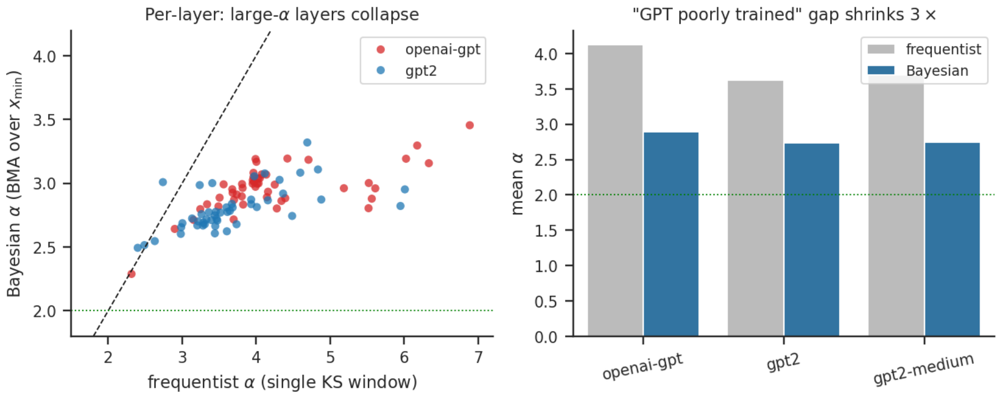

Figure 1: The GPT/GPT2 "training quality" gap is largely a scaling-window artifact. Left: each point is one weight matrix; the frequentist α (single KS window) versus the wwjd posterior α (Bayesian model-averaged over $x_{\min}$). Layers the point estimate calls "unusually large" ($\alpha > 5$) collapse toward $\alpha \approx 3$. Right: mean α per model; the frequentist gap that underwrites "GPT is poorly trained" shrinks 3.1× under the posterior.

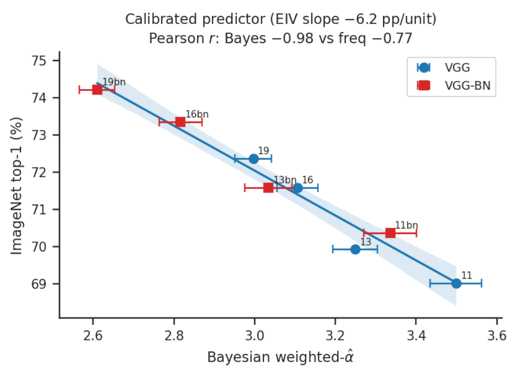

Figure 2: VGG weighted-$\hat{\alpha}$ as a calibrated accuracy predictor. The errors-in-variables regression propagates per-model $\hat{\alpha}$ posterior uncertainty (horizontal error bars); the band is the 95% predictive interval. The posterior $\hat{\alpha}$ tracks accuracy ($r = -0.98$) more strongly than the frequentist ($r = -0.77$).

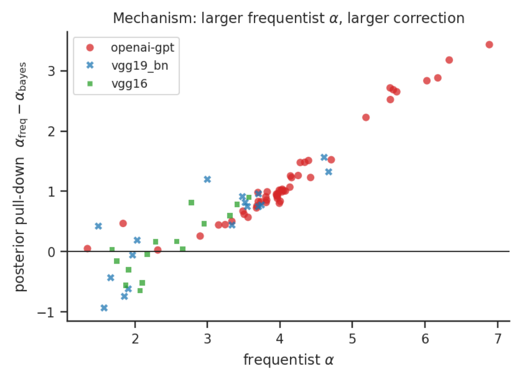

Figure 3: The unifying mechanism. The posterior pull-down ($\alpha_{\text{freq}} - \alpha_{\text{bayes}}$) grows with the frequentist α across image and language models: the single-window estimate inflates α most on the hard-to-fit (ambiguous-$x_{\min}$, truncated) layers, and the Bayesian treatment corrects it there; clean low-α layers barely move.

frequentist point estimate (Pearson $r = -0.77$) and find the Bayesian $\hat{\alpha}$ tracks accuracy more strongly, $r = -0.98$ (Figure 2): the model-averaging over $x_{\min}$ pulls down the inflated large-α layers, making the metric more predictive, not less. The metric must be the per-layer mean: the unweighted mean α gives $r = +0.79$, the wrong sign—a depth-confounded Simpson's paradox (Martin & Mahoney, 2021b)—which the weighting corrects and the posterior recovers cleanly. We close the loop with an errors-in-variables regression of accuracy on $\hat{\alpha}$ that propagates the per-model $\hat{\alpha}$ posterior uncertainty (Section 3): the posterior slope is -6.2 percentage points of top-1 accuracy per unit $\hat{\alpha}$ (95% credible interval $[-7.8, -4.8]$, $P(\text{slope} < 0) = 1.00$), residual $\sigma = 0.31$pp—the weights-only spectrum predicts ImageNet top-1 to within a third of a point. The measurement-error slope is slightly steeper than the naive ordinary-least-squares slope (-6.0), the expected attenuation correction.

Batch normalization breaks the power law in a way the point estimate hides. Across the BatchNorm VGG variants, the Bayesian model comparison rejects the power law (in favor

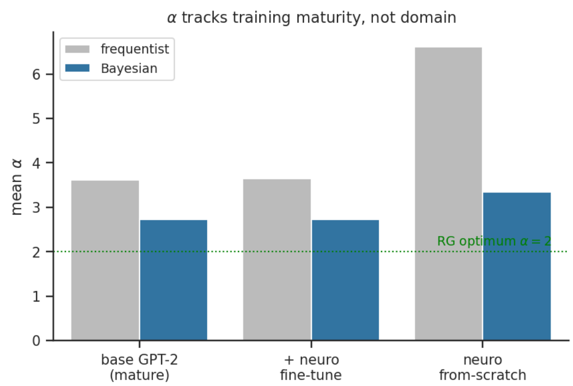

Figure 4: Controlled training-maturity test on GPT-2-124M we trained ourselves. A one-epoch neuroscience fine-tune of stock GPT-2 leaves mean α unchanged (mature spectrum, domain adaptation does not reshape it); a one-epoch from-scratch model sits well above the RG optimum $\alpha = 2$ (immature: the heavy tail has not formed). α tracks training maturity, not data domain.

of exponential or lognormal) for 8–17% of layers, rising with depth; plain VGG and all GPT models reject for 0%. The frequentist α, which never tests the power-law assumption, is blind to this structure.

α tracks training maturity, not data domain. The frozen-checkpoint results above compare models trained by others. To isolate what training itself does to the spectrum, we trained GPT-2-124M ourselves on an open neuroscience corpus (the 1.3B-token BrainGPT PMC subset) in two regimes and compared the spectra to stock GPT-2 (Figure 4). A gentle domain fine-tune (the pretrained model continued for one epoch, learning rate 2×10^{-5}) leaves the spectrum essentially untouched: mean posterior α moves from 2.73 to 2.73 (mean $|\alpha - 2|$ identical at 0.80). A model trained from scratch for the same one epoch is markedly less mature: posterior $\alpha = 3.35$ (frequentist 6.61), with mean $|\alpha - 2| = 1.35$ versus 0.80 for the pretrained base—the embedding and early attention/MLP layers carry the largest gap. A larger α is a lighter tail: the heavy-tailed structure has not yet formed. Training pulls α down toward the RG-optimal 2, and adaptation of an already-mature model does not move it—so the spectrum reflects how well-trained a model is rather than what it was trained on, a direct controlled confirmation of the HT-SR maturity claim. (Notably, even the one-epoch from-scratch model is still power-law-distributed: the Bayes factor rejects the power law for 0% of its layers, so immaturity manifests as a lighter power-law tail, not as loss of power-law structure.)

Watching the heavy tail form: the α trajectory. The from-scratch checkpoint above is a single immature snapshot because its learning rate (2×10^{-5}) was held to the fine-tune recipe. To resolve the dynamics, we trained a second GPT-2-124M from scratch with a proper from-scratch schedule (learning rate 6×10^{-4} cosine, $\beta_2 = 0.95$, weight decay 0.1), saving 17 checkpoints on a log-spaced schedule from random initialization to step 8837 (one epoch), and ran wwjd on every checkpoint (Figure 5). The mean posterior α falls from 3.69 at random initialization to 2.56 by step 1500—most of the descent is complete within the first $\sim$15% of one epoch—then relaxes to 2.70, essentially the stock-GPT-2 value (2.73): with a real learning rate, one epoch does reach maturity. The heavy tail forms layer-by-layer, with structure: token embeddings start lightest ($\alpha = 6.1$, nearest to pure noise) and travel farthest, while the attention and MLP output projections form their heavy tail first, undershooting $\alpha < 2.3$ near step 500 before relaxing back toward the band. The frequentist single-window estimate tells the same story far more crudely: it starts at 11.2, because the random-init Marchenko–Pastur bulk has no power-law tail and so no stable scaling window for a single KS-optimal $x_{\min}$ to find. The resulting posterior pull-down is 7.5 at initialization and collapses to 0.7 as the spectrum matures (Figure 5, right)—the identical mechanism

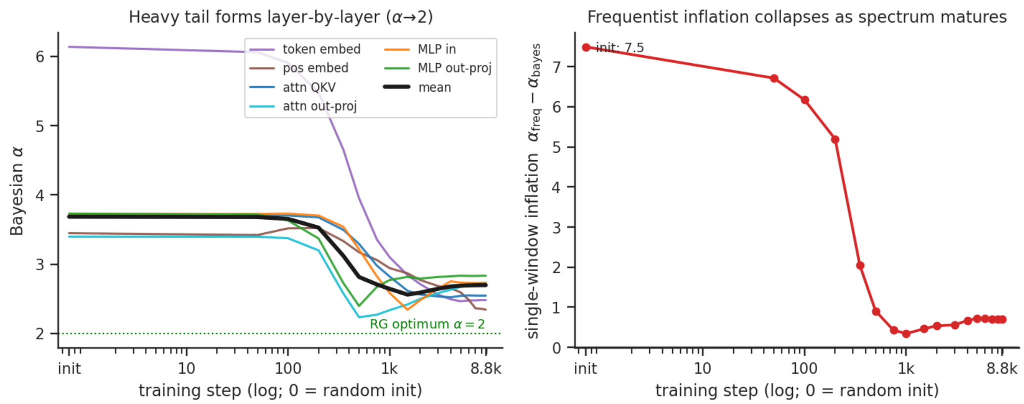

Figure 5: The HT-SR power law forming during training (GPT-2-124M from scratch, lr 6×10^{-4}, one epoch, wwjd on 17 checkpoints). Left: the Bayesian α of each layer type descends from the random-init light tail toward the RG optimum $\alpha = 2$; thick line is the mean ($3.69 \rightarrow 2.70$). Token embeddings travel farthest; output projections mature first. Right: the frequentist single-window inflation ($\alpha_{\text{freq}} - \alpha_{\text{bayes}}$) collapses from 7.5 at random initialization to 0.7—the Figure 3 mechanism in training time.

as the frozen-checkpoint plot (Figure 3), now resolved in training time: the single-window estimate is least trustworthy exactly where the scaling window is least defined, and the Bayesian treatment is most needed there.

The heavy tail is the weight-space shadow of a forming circuit. A spectral exponent matters only if it corresponds to computation. We tested this directly by running two mechanistic-interpretability probes on the same 17 checkpoints: the per-head induction score (Olsson et al., 2022) (does a head attend to, and copy, the token that followed the current token's previous occurrence?) and the OV copying score (Elhage et al., 2021)—the fraction of a head's output–value circuit eigenvalues with positive real part, an eigenvalue statistic in the same language as α. Three facts result (Figure 6). (i) Heavy-tail formation and circuit formation are locked. Across the trajectory the mean posterior α correlates -0.78 with the maximum induction score and $+0.97$ with the mean weight stable rank ($\|W\|_F^2/\|W\|_2^2$): the HT-SR exponent and the elementary stable rank track the same spectral concentration, and both anti-track the induction circuit. (ii) The spectrum leads the circuit. The stable rank collapses first—the spike separating from the Marchenko–Pastur bulk, reaching < 60 by step 500—and only then does an induction head ignite (score > 0.1 at step 750, > 0.4 at step 1000) while the top singular subspaces freeze (principal-angle drift $\rightarrow 0$). The heavy tail forms, and the computation comes online inside it $\sim$250–500 steps later. (iii) One circuit, two signatures. The dominant induction head (layer 7) acquires its behavioural induction score and its OV copying eigenvalues together, ending at induction 0.77 with 7 induction heads model-wide. This is a controlled, label-free demonstration that α is not a bystander statistic: the same steps that pull α toward 2 are the steps that build an interpretable circuit, and the spectral signature slightly precedes the behavioural one—consistent with the heavy tail being the enabling cause rather than a downstream correlate.

5 Cross-Modality Application: Where $\alpha = 2$ Pays Off, and Where It Does Not

The reanalysis above validates the diagnostics on models with known answers. We now turn them on the open question: does proximity to $\alpha = 2$ actually confer robustness, across modalities? We summarize a positive neuroimaging result and a negative foundation-model-scale pathology result.

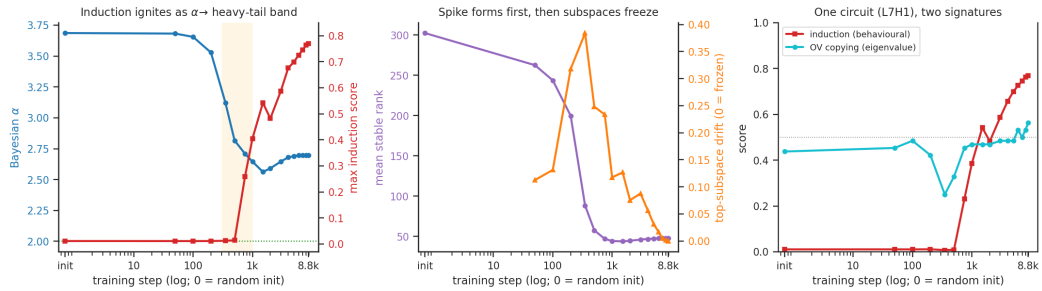

Figure 6: Mechanistic interpretability paired with heavy-tail formation, on the same from-scratch GPT-2 checkpoints. Left: the induction circuit ignites (red) just as the Bayesian α (blue) crosses into the heavy-tailed band (shaded: the spike-formation window). Center: the weight stable rank collapses first (purple, the spike forming) and the top singular subspaces then freeze (orange, drift $\to$ 0). Right: a single head (L7H1) comes online with both signatures at once—behavioural induction (red) and its OV copying eigenvalue fraction (cyan; 0.5 = no copying). α correlates -0.78 with induction and $+0.97$ with stable rank across the trajectory.

Table 1: Bayesian reanalysis of WeightWatcher's canonical checkpoints (auto-generated from data by paper/generate.py). $\bar{\alpha}$ is the mean per-layer exponent; "freq" is the single-$x_{\min}$ WeightWatcher estimate, "bayes" the wwjd posterior; PL-rej. is the fraction of layers the Bayes factor rejects as non-power-law.

Model	L	$\bar{\alpha}_{\text{freq}}$	$\bar{\alpha}_{\text{bayes}}$	CI width	PL-rej.	top-1
vgg11	10	2.34	2.30	0.49	0%	69.0
vgg13	12	2.34	2.28	0.51	0%	69.9
vgg16	15	2.48	2.36	0.56	0%	71.6
vgg19	18	2.71	2.42	0.60	0%	72.4
vgg11_bn	10	2.65	2.41	0.55	10%	70.4
vgg13_bn	12	2.55	2.37	0.58	8%	71.6
vgg16_bn	15	2.75	2.44	0.64	13%	73.4
vgg19_bn	18	2.93	2.42	0.61	17%	74.2
openai-gpt	50	4.12	2.90	0.54	0%	–
gpt2	50	3.62	2.73	0.49	0%	–
gpt2-medium	98	3.70	2.75	0.42	0%	–

Neuroimaging (positive, partial). A cross-modality survey of five neuroimaging checkpoints places the BrainIAC lineage near the RG optimum (mean $|\alpha - 2| \in \{0.12, 0.40\}$) and the MindEye lineage far from it ($\{3.49, 4.73\}$); the Vuong/Bayesian comparison formally rejects the power law for the MindEye per-subject layers with $\alpha \in [8, 11]$, a silent failure of bare α reporting. In a paired cohort sweep over 20 WAND subjects under TorchIO perturbations, the spectrally-closer checkpoint shows 3× lower CLS-embedding drift under bias field (Wilcoxon $p = 0.00022$, 9/10 perturbation seeds $p < 0.05$), with the effect concentrated in smooth, physics-realistic perturbations (bias field, motion) and absent for high-frequency artifacts (spike, noise)—the pattern the RG theory predicts. A controlled MLP ablation confirms the causal direction: alpha_loss drives mean α to 2.00 within 1000 steps and uniformly improves noise robustness.

Pathology (negative). The same alpha_loss regularizer, applied at foundation-model scale during continual DINOv2 pretraining of a ViT-S pathology encoder (10^{18}-FLOP budget), is robustness-neutral: pulling backbone weight-matrix α toward 2 leaves the downstream robustness probe unchanged and the mean probe score statistically indistinguishable from the unregularized recipe. The MLP-scale robustness gain does not transfer to a real foundation model in this modality. We report this as an honest boundary on the $\alpha = 2$ prediction: the diagnostic value of α (and of the Bayesian corrections above) is robust, but the interventional robustness payoff is scale- and modality-dependent.

6 Discussion

A single thread runs through the reanalysis: the frequentist single-window estimate inflates α on hard-to-fit layers, and the larger the frequentist α, the larger the posterior correction, because those are exactly the layers where the scaling window is ambiguous (a large share of their α variance is the window choice) and where the plain power law is mis-specified relative to a truncated one. Clean, near-optimal layers barely move. This is why a published conclusion built on large point-α layers—"GPT is poorly trained"—can weaken or dissolve under marginalization, while a conclusion built on the depth trend of well-behaved layers—the VGG accuracy relationship—survives and even sharpens. The practical recommendation is to report α with a credible interval, a power-law validity check, and, where the conclusion is comparative, the window-choice variance share; wwjd makes all three one function call.

Limitations. The Laplace evidences for the truncated power law and generalized Pareto are approximations; the cross-modality negative result is a single intervention; and the neuroimaging positive result, while controlled, spans one architecture family. Predictive model comparison (PSIS-LOO) and a mixture model separating the Marchenko–Pastur bulk from the heavy tail are natural next additions.

7 Conclusion

Heavy-tailed self-regularization is a powerful, falsifiable theory, and exactly because it is falsifiable it deserves uncertainty-aware estimation. We showed that a calibrated Bayesian treatment of the power-law exponent—posteriors, window marginalization, model comparison, and goodness-of-fit—changes conclusions drawn from the reference tool on its own canonical examples, turns the quality metric into a calibrated accuracy predictor, and draws an honest map of where the $\alpha = 2$ robustness prediction transfers across modalities. We release wwj/wwjd so that spectral diagnostics can be reported with error bars.

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Supplementary Material

A Reproducibility

Every number, table, and figure in Section 4 is generated from the per-layer analysis CSVs by paper/generate.py, which emits the headline macros (generated_values.tex), the LaTeX tables (generated_tables.tex), and the seaborn figures; bash paper/build.sh reweaves the manuscript from the data. The analysis itself is benchmarks/ww_replication.py (frequentist + Bayesian per layer), ww_gpt_detail.py (the GPT goodness-of-fit detail), and ww_bayes_diagnostics.py (EIV, TPL/GPD, ROPE, prior-sensitivity), all on the public checkpoints; wwj/wwjd carries a passing test suite.

B wwjd Implementation Details

The conjugate posterior is exact: writing $t = \log(\lambda/x_{\min})$, the Pareto tail becomes $\mathrm{Exp}(\beta)$ with $\beta = \alpha - 1$, and a $\mathrm{Gamma}(a_0, b_0)$ prior gives $\beta \mid \text{data} \sim \mathrm{Gamma}(a_0 + n, b_0 + \sum t)$; we use weakly informative $a_0 = b_0 = 10^{-3}$ so the flat-prior limit equals the CSN MLE. The BMA over $x_{\min}$ weights each candidate window by a base-measure-corrected marginal likelihood and reports a within-window/across-window variance split. The truncated-power-law and generalized-Pareto evidences use a Newton+Laplace approximation; all other evidences are closed form. The hierarchical model is a NumPyro partial-pooling fit; the EIV regression is a NumPyro measurement-error model standardized for sampling stability.

C GPT Large-α Layer Detail

Table 2 lists, for the GPT layers with frequentist $\alpha > 5$, the frequentist and posterior α with credible interval, the share of the α variance attributable to the $x_{\min}$-window choice, the power-law-vs-exponential log Bayes factor, and the posterior-predictive p-value.

Table 2: GPT (openai-gpt) layers with frequentist $\alpha > 5$. Auto-generated. $\alpha_{\mathrm{f}}/\alpha_{\mathrm{b}}$: frequentist/Bayesian exponent; win.var: $x_{\min}$-window variance share; $\mathrm{logBF}_{\mathrm{exp}}$: power-law vs exponential; ppc p: posterior-predictive.

shape	α_{f}	α_{b}	CI lo	CI hi	win.var	$\mathrm{logBF}_{\mathrm{exp}}$	ppc p
768x768	6.89	3.46	3.06	3.89	0.26	147.27	0.57
768x768	6.34	3.16	2.81	3.54	0.23	155.22	0.26
768x768	6.18	3.30	2.93	3.71	0.26	139.66	0.68
768x768	6.03	3.19	2.82	3.61	0.33	149.09	0.65
768x768	5.61	2.96	2.64	3.31	0.24	149.99	0.55
768x768	5.56	2.88	2.59	3.24	0.29	133.75	0.40
768x768	5.52	3.00	2.66	3.39	0.36	120.76	0.62
768x768	5.52	2.80	2.53	3.11	0.18	130.73	0.34
768x768	5.19	2.96	2.66	3.29	0.21	163.33	0.44

D Neuroimaging Experiment Details

The cross-modality survey covers five checkpoints (BrainIAC and a brain-age fine-tune (Pati et al., 2024); the realtime-MindEye decoder and two per-subject fine-tunes (Scotti et al., 2024)). The cohort sweep uses 20 WAND subjects, TorchIO (Pérez-García et al., 2021) perturbations (motion, bias field, ghosting, spike, noise) at five severities, with relative CLS-embedding drift as the metric and a paired Wilcoxon test across 10 perturbation seeds. The MLP ablation trains two 3-hidden-layer MLPs on sklearn digits (batch 128, 2000 steps, AdamW 10^{-3}) differing only in alpha_loss($\lambda = 0.01$, target 2). The pathology intervention applies alpha_loss during continual DINOv2 pretraining of a ViT-S encoder at a 10^{18}-FLOP budget.